

IMO — Innovative Maths Olympiad

Pattern of Questions, Syllabus and Marking Scheme · Grades 3 to 10

Grade 3

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

Number names (3-digit and 4-digit), comparing and forming numbers, and basic arithmetic operations (+, -, ×, ÷)

Measurement

Length, area, mass, volume, temperature, time, and money

Geometry

Shapes, patterns, and lines

Data Handling

Reading pictographs, tally marks, and bar graphs

Fractions

Understanding and identifying fractions

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marking for incorrect answers.
4. Students must use a blue or black ball-point pen to mark their answers on the OMR sheet.
5. The use of electronic gadgets, including calculators, mobile phones, and smartwatches, is strictly prohibited.

Grade 4

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Number Sense

5-digit and 6-digit number names, comparing and forming numbers, basic arithmetic operations, factors and multiples

Arithmetic and Measurement

Fractions and decimals, time and money, and the measurement of length, area, mass, and volume

Geometry and Data

Plane geometry, symmetry, and data handling

Instructions

1. This question paper contains 35 compulsory questions.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks, totaling 70 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen on the OMR sheet.
5. Electronic devices such as calculators, mobile phones, smartwatches, and log tables are strictly prohibited.

Grade 5

Total Questions: 35

Time: 1 hr

Total Marks: 70

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

7-digit and 8-digit numbers, patterns, and formation · Basic operations (+, -, ×, ÷), factors, and multiples

Fractions and Decimals

Fractions and decimals

Geometry

Geometry, shapes, and patterns

Measurement

Length, mass, volume, time, speed, and temperature · Money, perimeter, area, and volume

Data Handling

Data handling

Instructions

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There is no negative marking for incorrect answers.
4. Students must use a blue or black ball-point pen for the OMR sheet.
5. Electronic devices such as calculators, mobile phones, smartwatches, and log tables are strictly prohibited.

Grade 6

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

Knowing Our Numbers, Whole Numbers, and Playing with Numbers · Integers, Fractions and Decimals

Geometry

Basic Geometrical Ideas and Understanding Elementary Shapes · Symmetry

Mensuration and Data

Data Handling, Perimeter, and Area

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen on an OMR sheet.
5. Electronic devices such as calculators, mobile phones, or smartwatches are strictly prohibited.

Grade 7

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers and Algebra

Integers, Fractions and Decimals, and Rational Numbers · Exponents and Powers, and Algebraic Expressions · Simple Equations

Data Handling

Data Handling

Geometry and Mensuration

Lines and Angles, Triangles and Their Properties, Congruence of Triangles · Comparing Quantities, Perimeter and Area, and Visualising Solid Shapes

Instructions

1. This question paper contains 50 compulsory questions.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen for the OMR sheet.
5. Electronic devices, including calculators, mobile phones, smart watches, and log tables, are strictly prohibited.

Grade 8

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Numbers

Rational Numbers, Playing with Numbers, and Exponents and Powers · Squares and Square Roots, Cubes and Cube Roots

Algebra

Algebraic Expressions, Identities, Factorisation, and Linear Equations in One Variable

Commercial Mathematics and Proportion

Direct and Inverse Proportion, Commercial Mathematics

Geometry and Mensuration

Visualising Solid Shapes and Mensuration

Instructions

1. This question paper contains 50 compulsory questions.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks, totaling 100 marks. There are no negative marks for incorrect answers.
4. Answers must be marked on an OMR sheet using a blue or black ball-point pen.
5. Electronic devices such as calculators, mobile phones, smartwatches, or log tables are strictly prohibited.

Grade 9

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Algebra and Number System

Number System and Polynomials · Coordinate Geometry and Linear Equations in Two Variables

Geometry

Euclid's Geometry, Lines and Angles, Triangles, Quadrilaterals, and Circles

Mensuration and Statistics

Heron's Formula, Surface Areas and Volumes, and Statistics

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.

Grade 10

Total Questions: 50

Time: 1.5 hr

Total Marks: 100

Marks per Question: 2

Negative Marking: No negative marks for wrong answers

Algebra

Real Numbers, Polynomials, Pair of Linear Equations, and Quadratic Equations · Progressions

Geometry

Triangles and Coordinate Geometry · Circles

Trigonometry

Introduction to Trigonometry and Applications of Trigonometry

Mensuration and Data

Surface Areas and Volumes, Statistics, and Probability

Instructions

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for incorrect answers.
4. Students must use a blue or black ball-point pen to mark their answers on an OMR sheet.
5. Electronic devices, including calculators, mobile phones, and smartwatches, are strictly prohibited.